Per-site ΔRMSE from adding disturbance — M3 (C+T) vs M4 (C+T+D)

Below 0 = D reduces the site's RMSE | bars coloured by relative disturbance (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (106 sites, 388 site-years) · deadtrees v4.1

ΔRMSE = RMSE(without D) − RMSE(with D)

	XGBoost (Optuna) · 12m	XGBoost (Optuna) · 24m	LightGBM (Optuna) · 12m	LightGBM (Optuna) · 24m
GPPsat (µmol m⁻² s⁻¹)	57% improved	55% improved	55% improved	58% improved
NEPmax (µmol m⁻² s⁻¹)	59% improved	65% improved	68% improved	66% improved
ETmax (mm d⁻¹)	64% improved	58% improved	62% improved	58% improved
WUE (g C mm⁻¹)	58% improved	55% improved	52% improved	58% improved

Relative disturbance (%)
site maximum, 500 m
breaks 12.4 | 23.9

60
40
20
0

Sites (sorted by ΔRMSE within each panel)